

Population and Evolutionary Analysis of SLC30A8 **in the Context of Type 2 Diabetes**

Introduction and Background

Type 2 diabetes mellitus (T2D) has emerged as one of the most significant global health burdens today, earning its characterization as a “modern preventable pandemic” due to its steadily rising prevalence (Singer et al., 2021). Unlike acute infectious outbreaks, the progression of T2D has been gradual, contributing to delays in disease recognition, underdiagnosis, and insufficient public health prioritization. These challenges underscore the need for continued scientific investigation into the biological mechanisms underlying T2D susceptibility, with the objective of improving prevention, diagnostics, and therapeutics.

T2D is a multifactorial metabolic disease characterized by chronic hyperglycemia caused by insulin resistance and progressive dysfunction of pancreatic β -cells. While lifestyle factors such as obesity, diet, and sedentary behavior contribute strongly to disease initiation and progression, decades of epidemiological and genetic research have demonstrated that T2D cannot be explained by lifestyle alone (Lyssenko & Laakso, 2013). The genetic architecture of T2D is complex and polygenic, involving many common variants of modest effect as well as rare variants that exert larger impacts on β -cell biology and insulin regulation (American Diabetes Association, 2019). Genome-wide association studies (GWAS) have identified hundreds of loci contributing to T2D susceptibility, with many mapped to genes involved in insulin secretion, glucose metabolism, and pancreatic endocrine function.

Among these loci, SLC30A8, which encodes the zinc transporter ZnT8, has emerged as one of the many genes associated with T2D. It is located on chromosome 8 (117,134,995 - 117,176,714) based on UCSC Genome Browser on Human (GRCh38/hg38). ZnT8 plays a central role in pancreatic β -cells by transporting zinc ions into insulin secretory granules, enabling insulin crystallization, storage, and regulated secretion. GWAS first highlighted the importance of SLC30A8 when strong associations were detected between its variants and T2D risk across diverse populations (Sladek et al., 2007).

The contribution of SLC30A8 variants to risk or protection appears to vary across populations, emphasizing the importance of population genetics in interpreting disease-associated alleles. For example, the rs13266634 variant is associated with elevated T2D susceptibility in certain ancestral groups, demonstrating that allele frequencies and effect sizes differ across populations (Mashal et al., 2021). In parallel, evolutionary conservation provides a complementary perspective for identifying functionally relevant regions within SLC30A8. Comparative genomics analyses, such as phyloP scoring, detect coding and regulatory regions that have been preserved across millions of years of vertebrate evolution. High conservation often indicates functional constraint, suggesting that variants in these regions are more likely to impact gene function or regulation, while less conserved regions may highlight lineage-specific adaptation or relaxed selective pressure (Christmas et al., 2023). Thus, integrating evolutionary conservation with population-level data enables a more nuanced evaluation of which variants are potentially functionally relevant.

Overall, these findings highlight SLC30A8 as a gene of biological and clinical significance, positioned at the intersection of β -cell physiology, population genetics, and evolutionary constraint. However, gaps remain in understanding how variation within SLC30A8 interacts with population structure and evolutionary pressures to shape T2D risk, and how expression patterns of T2D-associated genes vary across populations. This study addresses these gaps by integrating three complementary analyses: population-level allele frequency variation of SLC30A8, evolutionary conservation of SLC30A8 variants, and population-level expression variation of T2D-associated genes. Together, these analyses provide a comprehensive framework for prioritizing functionally relevant variants and understanding their implications for T2D susceptibility in diverse populations.

Results

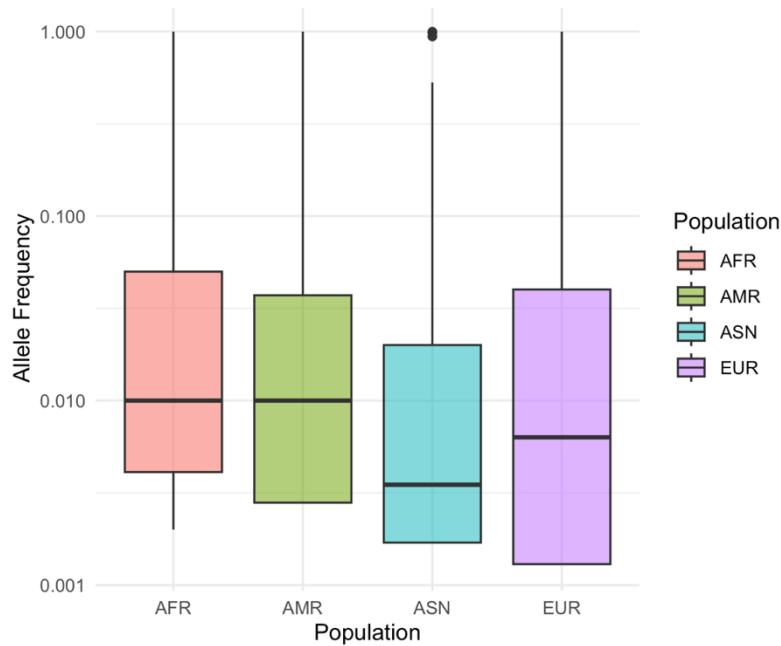
Population-Level Variation of SLC30A8

The allele frequencies of SLC30A8 variants were analyzed across four population groups. Population-specific analysis revealed that European (EUR) populations exhibited the highest average allele frequency, followed by East Asian (ASN), Admixed American (AMR), and African (AFR) populations (Table 1). Boxplot visualizations across populations were broadly similar, in which the visualization of allele frequency distributions indicates a right-skewed pattern across populations with a few common variants contributing disproportionately to the allele pool (Fig. 1).

Table 1: Average Allele Frequency Per Population

Population	AFR	AMR	ASN	EUR
Avg AF	0.0713	0.0870	0.102	0.112

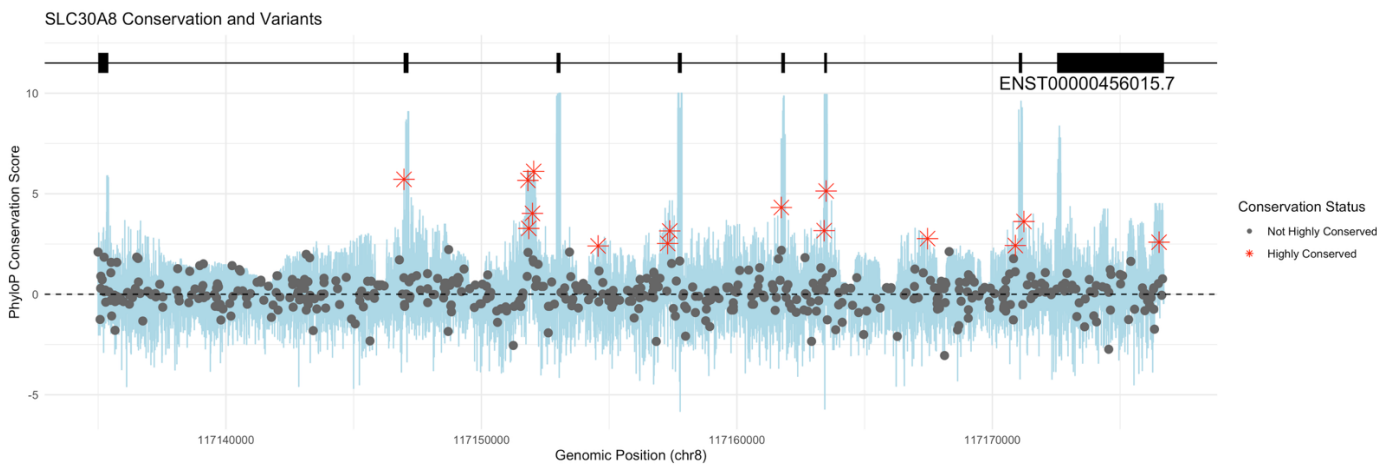
Figure 1: Boxplots of Allele Frequency Distributions Across Populations



Evolutionary Conservation of SLC30A8 Variants

PhyloP analysis identified approximately eight peaks of high evolutionary conservation, mostly corresponding to exons, and 1 peak of high evolutionary conservation on an intron region (Fig. 2). Fifteen variants fell within these conserved regions and are highlighted as red, starred datapoints: rs60737857, rs149563678, rs2316314, rs185665174, rs13258398, rs147579929, rs189016359, rs181436772, rs113210850, rs189887149, rs141153243, rs114167594, rs180728642, rs78075209, and rs77030255. When the highly conserved variants were cross-referenced against dbSNP and GWAS for clinical significance, none were annotated with known associations to type 2 diabetes or other diseases.

Figure 2: SLC30A8 Variants in Relation to PhyloP Conservation and Gene Structure



Population-Level Expression Variation for T2D Genes

Analysis of GEUVADIS gene expression data for T2D-associated genes identified several genes with high population-level variance. The top three genes with the greatest expression variance were GUCY1B1 (Variance = 0.538), LINC02506 (0.508), and ROBO1 (0.496) (Table 2). In contrast, SLC30A8 was ranked 101 as it showed a much lower variance value (0.076), indicating minimal expression differences across populations (Table 2).

Table 2: Population-Level Expression Variation of Top Three T2D-associated Genes and SLC30A8 Gene

Ranking	Gene Name	p-value	CEU	FIN	GBR	TSI	YRI	Variance
1	GUCY1B1	4E-09	4.05	2.61	4.45	3.67	4.28	0.538
2	LINC02506	5E-07	2.09	0.78	0.67	0.34	0.40	0.508
3	ROBO1	1E-08	2.859	1.37	2.00	1.04	1.52	0.496
...	...	...	...	...	...	...	...	...
101	SLC30A8	4E-67	0.269	0.80	0.20	0.26	0.68	0.076

Table 2 shows the top three T2D-associated genes and SLC30A8 with the highest variance in expression across five populations, based on the Top_T2D_Genes_Population_Expression_Variation dataset. The five populations are CEU (Utah residents with Northern and Western European ancestry), FIN (Finnish in Finland), GBR (British in England and Scotland), TSI (Toscani in Italy), and YRI (Yoruba in Ibadan, Nigeria). Columns include gene ID, gene name, p-value for population-level expression differences, expression values in the five populations, and variance of the expression values.

Discussion

The analyses provide a comprehensive view of population-level variation and evolutionary conservation for SLC30A8 and other T2D-associated genes.

According to the calculated average allele frequency of SLC30A8 variants of each population (Table 1) and the boxplot visualizations of the allele frequency distribution of SLC30A8 variant across each population (Fig. 2), most SLC30A8 variants are rare, with only model differences in mean allele frequency and are not population-specific. The low frequency of most variants likely reflects functional constraints on SLC30A8, a gene critical for insulin secretion, and suggests that purifying selection has maintained variant conservation across populations (Flannick et al., 2014). Because SLC30A8 plays an essential role in maintaining proper insulin granule function, variants that significantly disrupt its activity are likely deleterious. Purifying selection would act to remove such harmful variants from the population, leading to globally low allele frequencies and minimal population-specific differences (Mashal et al., 2021). This pattern suggests that even though some SLC30A8 variants contribute to T2D risk, these risk alleles are rare and conserved across populations. Furthermore, the similarity of allele distributions across populations may imply that environmental or genetic modifiers, rather than population-specific SLC30A8 variants, may play a larger role in observed differences in T2D prevalence.

PhyloP scores, which measure evolutionary conservation, support the functional importance of SLC30A8. Positive PhyloP scores indicate strong evolutionary constraint, highlighting regions where the encoded function is vital and shared across organisms, whereas regions with negative scores tend to reflect rapid evolution or lineage-specific adaptations (Christmas et al., 2023). In figure 2, eight peaks of high conservation were identified, primarily in exons. Exons encode the protein sequence and are often highly conserved because the functions they encode are essential and maintained across species. Changes in these regions are often deleterious, which explains the observed exonic peaks (Christmas et al., 2023). However, the fifteen SLC30A8 highly conserved variants that were highlighted in figure 2 were identified and cross-referenced with the dbSNP and GWAS database. The cross-reference to these databases identified these variants as intronic variations and showed no known clinical associations with T2D or other diseases. This is consistent with their intronic location since conservation in intronic regions often points to regulatory roles, such as splicing control or enhancer activity. Thus, despite the lack of documented clinical impact, the presence of variants in highly conserved regions suggests that these regions are functionally important and may warrant further investigation for potential regulatory or protective roles in β -cell physiology (Christmas et al., 2023).

Population-level expression analysis of T2D-associated genes revealed substantial variability among several loci. Genes, such as GUCY1B1, LINC02506, and ROBO1, showed high expression variance (approximately 0.5) across CEU, FIN, GBR, TSI, and YRI populations, indicating that their expression levels differ markedly among groups and may reflect population-specific regulatory differences (Table 2). On the other hand, SLC30A8 ranked 101st, displaying a low expression variance of only 0.076 (Table 2). This low variance indicates that SLC30A8 expression is highly consistent across populations, suggesting strong regulatory constraint (Alemu et al., 2014). Genes with low expression variance are often those with essential biological functions because both overexpression and under-expression can disrupt critical cellular processes. In the case of SLC30A8, which is required for zinc transport in pancreatic β -cells, maintaining stable expression likely ensures proper insulin processing and secretion.

Importantly, the three analyses consistently demonstrate that SLC30A8 is a highly conserved gene. Its variants are globally rare, its evolutionarily constrained regions are largely preserved across species, and its expression levels remain stable across diverse human populations. These converging results indicate that SLC30A8 is under strong purifying selection, reflecting its essential role in β -cell physiology and insulin secretion. While SLC30A8 is a well-established T2D risk gene, this study shows that its contribution to disease is unlikely to stem from common population-specific differences but rather from rare functional variants or regulatory mechanisms that remain to be fully explored.

Several limitations should be considered when interpreting these findings. First, the allele frequency data relied on publicly available population databases, which may underrepresent rare variants or populations with limited sampling. Similarly, the PhyloP conservation analysis highlights functionally constrained regions but cannot directly determine whether individual variants have regulatory or phenotypic effect as many conserved intronic variants remain functionally untested. The expression variance analysis also introduces limitations, as bulk expression estimates can obscure cell-type-specific regulation, especially for genes like SLC30A8, which is expressed predominantly in pancreatic β -cells. Additionally, cross-population comparisons may be affected by differences in sample size, tissue availability, or technical batch effects, all of which may contribute to variability unrelated to true biological differences.

This project deviated from the original proposal, which aimed to examine allele frequencies and evolutionary conservation of only the SLC30A8 gene to uncover population-specific patterns and selective pressures. During analysis, however, it became clear that SLC30A8 shows very little variance across populations, limiting the ability to draw meaningful conclusions about regulatory divergence. Because of this, the final project expanded to include expression variance across multiple T2D-associated genes, which provided more interpretable contrasts among populations. This shift broadened the scope beyond SLC30A8's evolutionary profile and toward understanding population-level regulator differences, but it also meant that the final analyses no longer strictly aligned with the initial gene-focused evolutionary framework.

Methods and Datasets

To analyze the population-level variation and evolutionary conservation of SLC30A8, variants were first extracted from the 1000 Genomes Project Phase 1 integrated call set (v3, 20101123) for chromosome 8. Using Linux command-line tools, variants within the genomic coordinates of SLC30A8 (117,134,995 – 117,176,714, GRCh38/hg38) were extracted and formatted into a BED file. Allele frequencies for each variant were extracted for four major populations: European (EUR), East Asian (ASN), Admixed American (AMR), and African (AFR). Average allele frequencies were calculated for each population, and population-specific distributions were visualized with ggplot2 as boxplots.

Evolutionary conservation was analyzed using PhyloP 100-way vertebrate conservation scores obtained from the UCSC Genome Browser. Base-wise conservation scores in the SLC30A8 region were mapped to variant coordinates, and highly conserved variants were defined as those with PhyloP scores above 2.27. Conservation patterns were visualized along the genomic coordinates with R, integrating variant positions and exon/intron structures (ENST0000456015.7 transcript) using scatterplots and line plots to show conservation across the gene.

Population-level expression variation for T2D-associated genes was analyzed using the GEUVADIS RNA-sequencing dataset, which contains FPKM expression values for lymphoblastoid cell lines from approximately 500 individuals of the 1000 Genomes Project. Sample information, including population identity, was obtained from the 1000 Genomes sample spreadsheet, representing five populations: CEU (Utah residents with Northern and Western European ancestry), FIN (Finnish in Finland), GBR (British in England and Scotland), TSI (Toscani in Italy), and YRI (Yoruba in Ibadan, Nigeria). For each gene, the mean expression was calculated within each population. Expression values were log2-transformed to stabilize variance, and population-level variance was computed across the populations. Finally, the expression matrix was filtered to retain only T2D-associated genes that were obtained from the GWAS catalog, and genes were ranked by descending population-level variance.

Table 3: Datasets/Databases Utilized

Dataset/ Database	Source	Version	Link
Phase 1 Integrated Call Set	1000 Genomes Project	V3 (20101123)	https://ftp-trace.ncbi.nih.gov/1000genomes/ftp/phase1/analysis_results/integrated_call_sets/ALL.chr8.integrated_phase1_v3.20101123.snps_indels_svs.genotypes.vcf.gz
PhyloP 100-way Vertebrate Conservation Scores	UCSC Genome Browser	GRCh38/hg38	Basewise evolutionary conservation scores across the SLC30A8 region, used to identify highly conserved variants
dbSNP	NCBI	Build 155 (2024)	Cross-reference highly conserved SLC30A8 variants for known clinical associations
GEUVADI RNA-Seq Dataset	EUVADIS Consortium	2013 Release	https://www.ebi.ac.uk/gxa/experiments/E-GEUV-1/Downloads?ref=biostudies
1000 Genomes sample information	1000 Genomes Project	2013 Release	ftp://ftp.1000genomes.ebi.ac.uk/vol1/ftp/technical/working/20130606_sample_info/20130606_sample_info.xlsx

Table 4: Datasets Created

Datasets	Link	Description
SLC30A8_variants.tsv	https://github.com/kluong4/Portfolio/blob/main/SLC30A8-T2D-analysis/Results/SLC30A8_variants.tsv	SLC30A8 variant data from Phase 1 Integrated Call Set, PhyloP 100-way Vertebrate Conservation Scores, and dbSNP.
Top_T2D_Genes_Population_Expression_Variation.tsv	https://github.com/kluong4/Portfolio/blob/main/SLC30A8-T2D-analysis/Results/Top_T2D_Genes_Population_Expression_Variation.tsv	T2D-associated genes from GEUVADI RNA-Seq dataset with average FPKM expression values for each population and sorted by population-level expression variation

References

- Alemu, E. Y., Carl, J. W., Corrada Bravo, H., & Hannenhalli, S. (2014). Determinants of expression variability. *Nucleic Acids Research*, 42(6), 3503–3514. <https://doi.org/10.1093/nar/gkt1364>
- American Diabetes Association. (2019). 2. Classification and diagnosis of diabetes: *Standards of medical care in diabetes—2019. Diabetes Care*, 42(Supplement_1), S13–S28. <https://doi.org/10.2337/dc19-S002>
- Christmas, M. J., Kaplow, I. M., Genereux, D. P., Dong, M. X., Hughes, G. M., Li, X., Sullivan, P. F., Hindle, A. G., Andrews, G., Armstrong, J. C., Bianchi, M., Breit, A. M., Diekhans, M., Fanter, C., Foley, N. M., Goodman, D. B., Goodman, L., Keough, K. C., Kirilenko, B., ... Karlsson, E. K. (2023). Evolutionary constraint and innovation across hundreds of placental mammals. *Science (New York, N.Y.)*, 380(6643), eabn3943. <https://doi.org/10.1126/science.abn3943>
- Flannick, J., Thorleifsson, G., Beer, N. L., Jacobs, S. B. R., Grarup, N., Burt, N. P., Mahajan, A., Fuchsberger, C., Atzmon, G., Benediktsson, R., Blangero, J., Bowden, D. W., Brandslund, I., Brosnan, J., Burslem, F., Chambers, J., Cho, Y. S., Christensen, C., Douglas, D. A., ... Altshuler, D. (2014). Loss-of-function mutations in SLC30A8 protect against type 2 diabetes. *Nature Genetics*, 46(4), 357–363. <https://doi.org/10.1038/ng.2915>
- Lyssenko, V., & Laakso, M. (2013). Genetic screening for the risk of type 2 diabetes. *Diabetes Care*, 36(Suppl 2), S120–S126. <https://doi.org/10.2337/dcS13-2009>
- Mashal, S., Khanfar, M., Al-Khalayfa, S., Srour, L., Mustafa, L., Hakooz, N. M., Zayed, A. A., Khader, Y. S., & Azab, B. (2021). SLC30A8 gene polymorphism rs13266634 associated with increased risk for developing type 2 diabetes mellitus in Jordanian population. *Gene*, 768, 145279. <https://doi.org/10.1016/j.gene.2020.145279>
- Singer, M. E., Dorrance, K. A., Oxenreiter, M. M., Yan, K. R., & Close, K. L. (2021). The type 2 diabetes ‘modern preventable pandemic’ and replicable lessons from the COVID-19 crisis. *Preventive Medicine Reports*, 25, 101636. <https://doi.org/10.1016/j.pmedr.2021.101636>
- Sladek, R., Rocheleau, G., Rung, J., Dina, C., Shen, L., Serre, D., Boutin, P., Vincent, D., Belisle, A., Hadjadj, S., Balkau, B., Heude, B., Charpentier, G., Hudson, T. J., Montpetit, A., Pshezhetsky, A. V., Prentki, M., Posner, B. I., Balding, D. J., ... Froguel, P. (2007). A genome-wide association study identifies novel risk loci for type 2 diabetes. *Nature*, 445(7130), 881–885. <https://doi.org/10.1038/nature05616>